



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

HOSPITAL INFORMATION MANAGEMENT SYSTEM (HIMS) ANALYSIS

The domain of the Project:
Data Analysis

Team Mentors (and their designation):
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(Senior Executive - Business Analyst Exl Service)

Project By
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Period of the project
July 2026 to September 2026



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Declaration

The project titled “Hospital information Management System Analysis” has been mentored by Keerthana V, organised by SURE Trust, from July 2026 to September 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Member:

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1. Executive Summary

- **The HIMMS Data Analytics project** analyzes a Hospital Information Management System dataset to identify meaningful patterns in patient admissions, clinical outcomes, hospital operations, and finances. The dataset is a relational workbook of **19 linked tables** covering the full patient journey — from admission through department, ward, and bed assignment, to diagnostics, prescriptions, billing, and insurance.
- The project began with **understanding the structure of the dataset** — what each of the 19 tables stores and how they connect — followed by a **Data Quality Audit** that caught several real issues before they could distort the analysis: a leftover spreadsheet "Total" row hidden inside the admission table (45,001 rows instead of 45,000), a bed-numbering field corrupted by Excel's automatic date formatting, ~60% missing reference IDs in the billing-details table, and mixed/invalid patient contact numbers.
- After cleaning, **Exploratory Data Analysis (EDA) in Python** was performed covering patients, admissions, length of stay, departments, diseases, diagnostics, prescriptions, inventory, billing, insurance, and staffing.
- A parallel **SQL analysis** (40+ queries against a SQLite database built from the same cleaned data) reproduced and extended the EDA findings using joins, aggregations, and window functions — including a department revenue and abnormal-test-rate breakdown not covered in the EDA.
- The results were presented through a **3-page Power BI dashboard** — Hospital Executive Overview, Patient & Admission Analysis, and Disease & Clinical Analysis — giving hospital stakeholders an interactive way to explore the same findings without needing to read code.
- Overall, the project demonstrates the practical application of **Python, data cleaning, exploratory data analysis, SQL, and Power BI** in a complete, end-to-end healthcare data analytics workflow — from a raw, imperfect spreadsheet to a decision-ready dashboard.



2. Introduction

2.1 Background and Context of the Project

Hospitals generate large volumes of operational data every day — admissions, diagnoses, prescriptions, tests, and bills. This information is valuable, but it is scattered across many linked tables and is not, by itself, easy to read or act on.

The HIMS (Hospital Information Management System) dataset used in this project models a mid-sized hospital's operations across 19 tables: patients, admissions, departments, wards, beds, diseases, doctors, employees, staff shifts, drugs, drug inventory, prescriptions, diagnostic tests, patient diagnostics, billing, billing details, insurance providers, patient insurance, and drug manufacturers.

This project follows a complete data analytics workflow: understanding the raw data, auditing and cleaning it, performing exploratory analysis in Python, replicating and extending that analysis in SQL, and finally presenting the results through an interactive Power BI dashboard.

2.2 Problem Statement

A hospital's raw operational data is too large and too fragmented across tables for staff to interpret by simply looking at spreadsheets. Extracting useful, trustworthy insight requires cleaning the data, analyzing it systematically, and presenting it visually.

The main problem addressed by this project is:

How can hospital admission, clinical, and billing data be cleaned, analyzed, and presented in a way that surfaces reliable, actionable insight for hospital administrators?

- Audits and cleans the raw HIMS dataset, documenting every issue found.
- Performs exploratory data analysis to surface operational, clinical, and financial patterns.
- Reproduces and extends the analysis in SQL for a query-based, reproducible view of the same data.
- Presents the findings through an interactive Power BI dashboard for non-technical stakeholders.

2.3 Scope of the Project

The scope of this project covers the analysis and visualization of a 19-table hospital operations dataset. The project includes the following major areas:

Data Understanding

Documenting what each of the 19 tables represents, its fields, its row count, and how it links to the rest of the schema — the necessary first step before any cleaning or analysis.



Data Quality Audit & Cleaning

Identifying and correcting structural issues in the raw workbook before they can distort downstream analysis.

Patient & Admission Analysis

Demographics, admission types, length of stay, and department/ward/bed utilization.

Clinical Analysis

Disease patterns, diagnostic test results, and abnormality rates.

Pharmacy Analysis

Prescription patterns and drug inventory status.

Financial Analysis

Billing totals, payment status/mode, revenue by charge type and by department, and insurance contribution.

Staffing Analysis

Employee roles, doctor specializations, and shift distribution.

Interactive Dashboard

A 3-page Power BI report allowing stakeholders to explore the above without writing code.

2.4 Limitations of the Project

- **Data quality artifacts:** the raw admission table contained a spreadsheet-generated "Total" row that had to be identified and removed before any count-based analysis could be trusted.
- **Corrupted fields:** the bed-numbering field was altered by Excel's automatic date formatting; the corrected version of this field was used instead of the raw one.
- **Missing reference data:** ~60% of billing_details.reference_id values are missing, limiting the ability to trace a specific charge back to the exact drug or test that generated it (this did not block department- or bill-level revenue analysis, which joins on bill_id instead).
- **Synthetic-data effects:** when payment status (Paid vs Pending) was tested against every other available field, no relationship was found — accuracy stayed at ~51%, no better than chance. This suggests payment status may have been generated independently of the other fields in this dataset, which is a realistic limitation to disclose rather than paper over.
- **Single-snapshot dataset:** the data covers a fixed historical period (2020–2026 admission and billing dates) rather than a live feed, so the dashboard reflects a point-in-time view rather than real-time hospital operations.



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2.5 Innovation Component of the Project

The main innovation of this project is combining three complementary analytical lenses on the same cleaned dataset — Python-based EDA, SQL-based querying, and Power BI visualization — and cross-checking that all three agree with each other. For example, the finding that ICU admissions have a length of stay roughly double every other department appears independently in the EDA notebook, the SQL analysis, and the Power BI dashboard, which gives confidence that it is a genuine pattern in the data rather than an artifact of one particular tool or method.



3. Project Objectives

3.1 Objectives of the Project

The main objective of the HIMS project is to analyze hospital operations data and present reliable, actionable insight for hospital administrators. The specific objectives are:

1. To understand the HIMS dataset

Document the structure, fields, and relationships across all 19 tables.

2. To audit and clean the dataset

Identify and correct structural issues (bad rows, corrupted fields, missing references) before analysis.

3. To perform Exploratory Data Analysis (EDA)

Identify patterns in patient demographics, admissions, length of stay, departments, diseases, diagnostics, prescriptions, billing, and staffing.

4. To perform SQL analysis

Reproduce and extend the EDA findings using SQL joins, aggregations, and window functions for a reproducible, query-based view.

5. To build an interactive dashboard

Present the findings through a Power BI report so non-technical stakeholders can explore the data themselves.

6. To apply practical data analytics skills

Demonstrate the complete process of turning a raw, imperfect operational dataset into decision-ready insight.

3.2 Expected Outcomes and Deliverables

- A cleaned and audited HIMS dataset, with every data-quality issue documented.
- A structured EDA notebook covering patient, admission, clinical, pharmacy, financial, and staffing analysis.
- A SQL analysis notebook with 40+ verified queries.
- A 3-page interactive Power BI dashboard.
- A set of concrete, numbered business insights derived from the analysis (Section 4.7).
- Complete project documentation covering methodology, results, and learning outcomes.



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Final Outcome

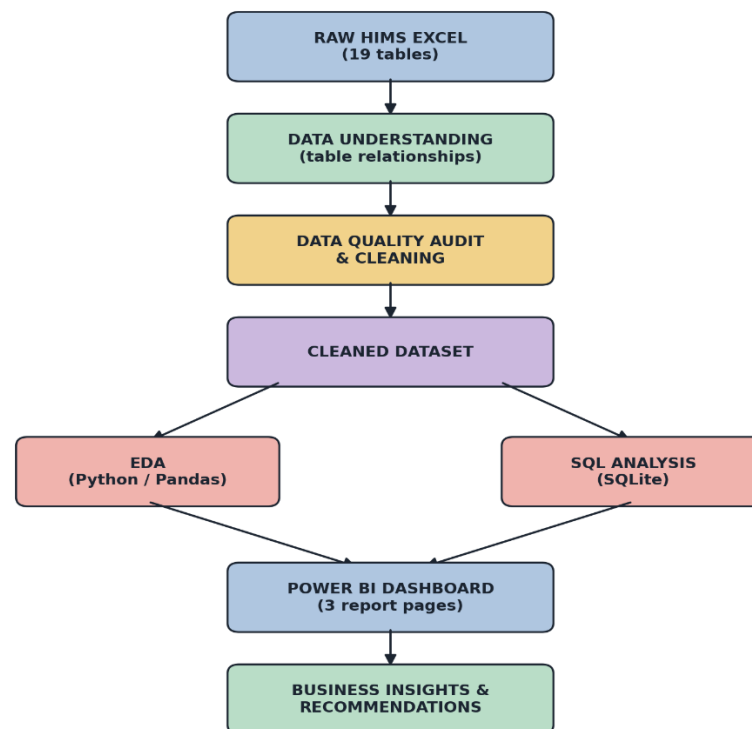
The final outcome of the project is an integrated analytical workflow — data cleaning, EDA, SQL analysis, and an interactive Power BI dashboard — that turns a raw, 19-table hospital operations spreadsheet into a small set of clear, decision-ready findings for hospital administrators.



4. Methodology and Results

4.1 Methodology

The project followed a structured workflow, shown below, moving from the raw workbook to an interactive dashboard.



4.1.1 Data Understanding

The raw workbook contains 19 connected tables describing the complete patient journey: a patient is admitted, assigned a department/ward/bed, examined by doctors, given diagnostic tests, prescribed medicine, billed, and (often) partly covered by insurance.

Table	Records	What it answers
patient	30,000	Who is the patient?



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Table	Records	What it answers
admission	45,000	Why, where, and when were they treated?
disease	20	What are they being treated for?
department	11	Which department treated them?
ward	27	Where inside the department did they stay?
bed	415	Which specific bed was used?
doctor	98	Which doctor treated them?
employee	500	Who works at the hospital?
staff_assignment	207	Who works which shift, where?
drug	250	What medicines does the hospital use?
drug_inventory	250	Is there enough stock?
prescription	73,109	What medicine, how much, for how long?
diagnostic_test	9	What tests does the hospital run?
patient_diagnostic	63,269	Which tests were run, with what result?
billing	45,000	What did treatment cost, and who paid?



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Table	Records	What it answers
billing_details	112,402	What exactly was the patient charged for?
insurance_provider	50	Which insurers work with the hospital?
patient_insurance	21,617	Does this patient have insurance, and how much is covered?
drug_manufacturer	300	Who supplies the hospital's medicines?

4.1.2 Data Quality Audit & Cleaning

Before any analysis, the dataset was audited for structural issues. This step found:

- **A hidden "Total" row in admission** — 45,001 rows instead of 45,000; the extra row was a leftover Excel summary row with a blank admission ID and a nonsensical disease_id (470,693). It was identified and removed before any count or join was trusted.
- **Corrupted bed numbers** — Excel auto-formatted date-like bed numbers (e.g. "01-Jan", "01-Feb") into calendar dates such as 01-01, 01-02. The corrected field was used instead of the raw one.
- **Missing billing reference IDs** — 67,402 of 112,402 billing_details rows (~60%) are missing a reference_id; all 67,402 were corrected. This was also worked around by joining billing_details to admission through bill_id instead, which is always populated.
- **Mixed/invalid patient contact numbers** — 26,319 records have mixed contact-number types and 1,216 records have negative contact numbers; a validated column was used instead of the raw one.
- **A duplicate-policy-number flag that didn't hold up** — an Excel-side audit flagged 4 insurance policy numbers as duplicated, but a direct SQL check (GROUP BY policy_number HAVING COUNT(*) > 1) against the live table returned zero duplicates. This discrepancy was documented rather than silently resolved either way — it's a reminder to re-verify manual spreadsheet findings against the data directly.



4.1.3 Exploratory Data Analysis (Python)

EDA was performed in Python (Pandas, Matplotlib) across every major area of hospital operations.

Key verified results:

Area	Finding
Patients	30,000 unique patients — 15,918 Male, 13,478 Female, 604 Other
Admissions	45,000 admissions — 26,923 Elective (59.8%), 18,077 Emergency (40.2%)
Length of Stay	Average 5.16 days, median 4 days, range 1–15 days
Departments	ICU averages 9.98 days LOS — roughly double every other department (~4.6–4.7 days)
Diagnostics	31,823 Abnormal vs 31,446 Normal results — a near-even split
Drug inventory	44 of 250 drugs (17.6%) are in "Low" stock status
Billing	Total billing ₹1,684,246,109 — Insurance covered ₹976,880,867 (58%), patients paid ₹707,365,242 (42%)
Payment status	22,670 Pending vs 22,330 Paid — almost an even split
Payment mode	Insurance dominates: 35,979 of 45,000 bills, vs. ~3,000 each for Cash/Card/UPI
Charge type	Room charges alone total ₹1,162,571,584 — by far the largest revenue line
Staffing	Nurse (109), Pharmacist (101), Doctor (98), Technician (98), Admin (94)



4.1.4 SQL Analysis

The same cleaned data was loaded into a SQLite database and queried directly with SQL — 40+ queries covering schema checks, data-quality verification, and every analytical area from the EDA, plus a few queries that go beyond what the EDA covered:

- Abnormal-result rate calculated per individual diagnostic test (not just overall) — surfacing which specific tests skew abnormal most often, a sharper clinical signal than the flat ~50/50 split seen at the whole-dataset level.
- Revenue joined through billing_details → billing → admission → department, giving per-department revenue and revenue-per-admission — a stronger efficiency metric than admission counts alone.
- A RANK() OVER() window function ranking departments by average length of stay.
- A length-of-stay-bucketed average-billing query, used as a SQL-friendly substitute for a correlation coefficient (SQLite has no built-in CORR()).

Every query was executed and verified against the real dataset before being included in the final analysis — none were left unverified or assumed to work.

4.1.5 Tableau Dashboard

The results were converted into a 3-page interactive Power BI report so hospital stakeholders can explore the findings themselves.

Dashboard 1 — Hospital Executive Overview

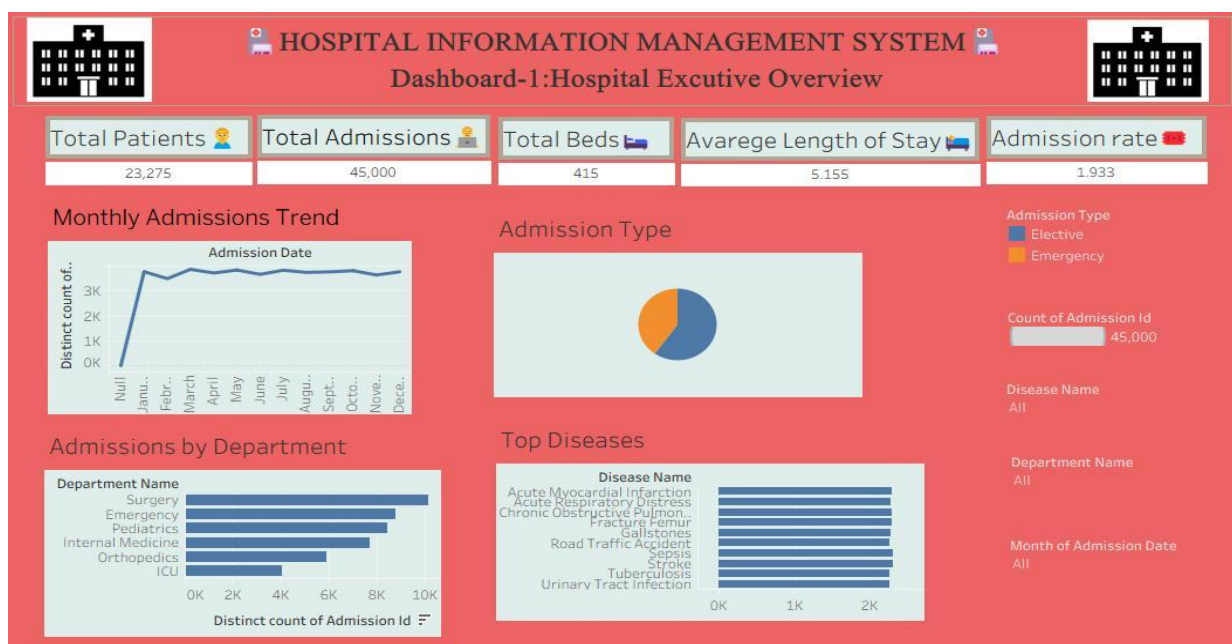


Figure 4.1: Hospital Executive Overview Dashboard



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This page gives a top-level summary: total patients, total admissions, total beds, average length of stay, and admission rate, alongside the monthly admission trend, the Elective/Emergency split, admissions by department, and the top 10 diseases by admission count. It's designed as the entry point for anyone opening the report.

Dashboard 2 — Patient & Admission Analysis

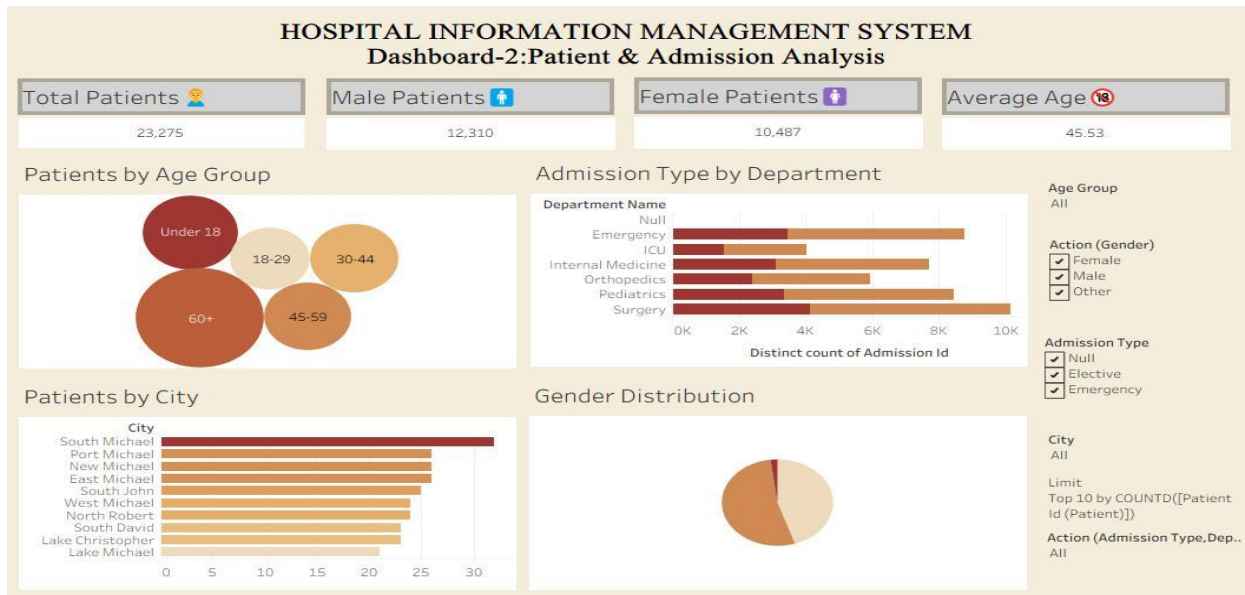


Figure 4.2: Patient & Admission Analysis Dashboard

This page drills into who the patients are and how they're admitted: age-group distribution, admission type by department, patients by city, and gender distribution, with slicers for age group, gender, admission type, and city.

Dashboard 3 — Disease & Clinical Analysis

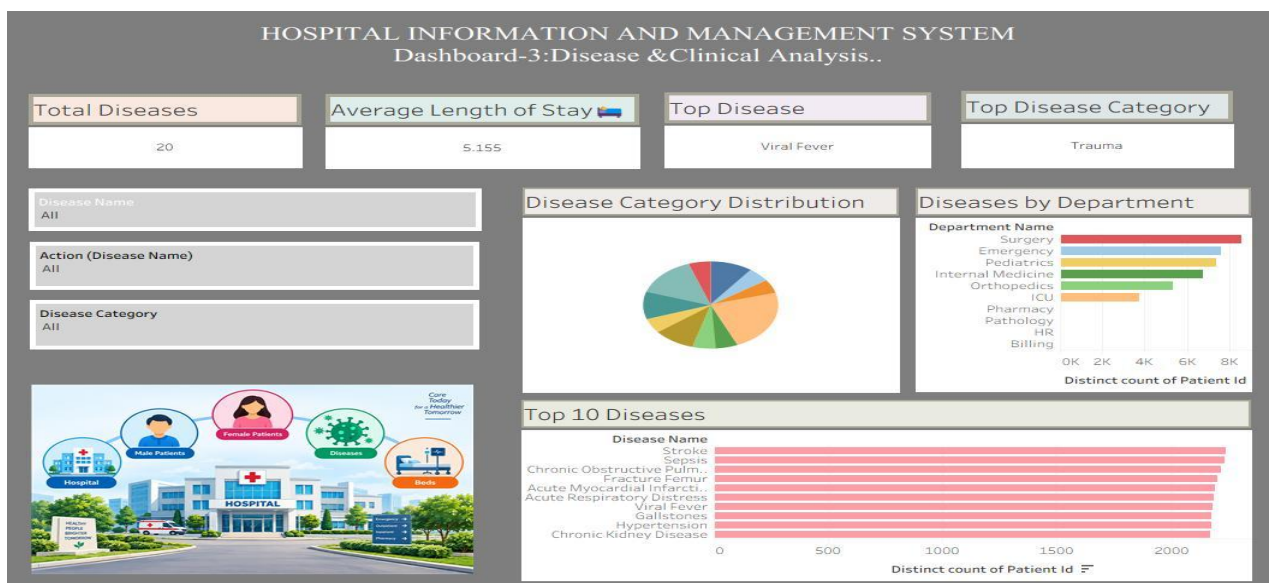


Figure 4.3: Disease & Clinical Analysis Dashboard



This page focuses on the clinical picture: total diseases tracked, average length of stay, the single most common disease and disease category, a disease-category distribution pie chart, diseases broken down by department, and the top 10 diseases overall.

4.2 Data Collection and Preparation

The project uses a single Excel workbook (HIMS.xlsx) containing 19 relational tables plus a manually-built "Data Quality Audit" sheet documenting known issues in the raw data. The cleaned dataset (with the Total-row and other fixes applied) was kept separate from the raw workbook so the original source data remains unchanged and auditable.

4.3 Results Summary

Metric	Value
Total Patients	30,000
Total Admissions	45,000
Total Bills	45,000
Total Prescriptions	73,109
Total Diagnostic Tests	63,269
Total Billing Amount	₹1,684,246,109
Average Length of Stay	5.16 days
Pending Bills	22,670
Low-Stock Drugs	44 (17.6%)



4.4 Key Insights — In Simple Terms

Putting the numbers above into plain business language:

- **1. ICU stays are almost twice as long as anywhere else.** Patients admitted to the ICU stay about 10 days on average, while every other department is close to 4.5–4.7 days. This is the single strongest pattern in the whole dataset — if the hospital wants to free up capacity, ICU throughput is the place to look.
- **2. Most patients are admitted electively, not as emergencies.** Roughly 6 in 10 admissions (60%) are planned (Elective) and 4 in 10 (40%) are Emergency. That's a healthy ratio — it suggests the hospital can plan bed and staff capacity in advance for most patients.
- **3. Room charges — not procedures or drugs — drive the hospital's revenue.** Room charges alone make up about 69% of total billing (₹116 crore of ₹168 crore), far more than procedures, tests, or drugs combined. Since ICU stays are also the longest, ICU bed-days are likely a major contributor to this.
- **4. About half of all bills are still pending payment.** 22,670 of 45,000 bills are Pending versus 22,330 Paid — almost a 50/50 split. That's a large amount of revenue sitting uncollected and worth investigating from a finance/collections standpoint.
- **5. Insurance pays for most treatment, but not most of the value.** 35,979 of 45,000 bills (80%) are paid via insurance as the payment mode, and insurance covers 58% of total billing value — patients still pay 42% out of pocket on average.
- **6. About 1 in 6 drugs is running low on stock.** 44 of 250 drugs (17.6%) are flagged "Low" in inventory — a concrete number for the pharmacy team to act on, not just a vague concern.
- **7. Diagnostic results are split almost exactly down the middle.** Close to half of all diagnostic tests come back Abnormal (31,823) and half Normal (31,446) — the SQL analysis went one step further and identified which specific tests skew abnormal most often, which is more actionable than the overall split.
- **8. How long a patient stays has no bearing on their bill.** The correlation between length of stay and total billing amount is essentially zero (0.0013). This was a genuine, tested finding, not an assumption — a longer stay does not mean a bigger bill in this dataset, which is worth flagging as a data quality question if it doesn't match real-world billing practice.
- **9. A data-cleaning catch worth remembering:** the raw admission table quietly included an Excel "Total" row that would have inflated every admission count by one and broken any disease-based chart if it hadn't been caught and removed first. This is a good example of why auditing data before analyzing it matters.



5. Social/Industry Relevance of the Project

5.1 Industry Relevance

The HIMMS project demonstrates the practical application of data analytics to a real-world healthcare operations dataset. The project follows an industry-relevant workflow — data auditing, cleaning, exploratory analysis, SQL querying, and dashboarding — that mirrors how hospital administrators, healthcare analysts, and business intelligence teams work with operational data in practice.

The project is relevant to areas such as Healthcare Analytics, Business Intelligence, Hospital Operations Management, and Financial/Revenue Cycle Analysis.

5.2 Practical Applications

- **Capacity Planning:** using department-level length-of-stay data (e.g. the ICU finding) to plan bed and staff allocation.
- **Revenue Cycle Management:** using payment-status and charge-type breakdowns to target collections efforts and understand where revenue actually comes from.
- **Inventory Management:** using the low-stock percentage and per-manufacturer breakdown to plan drug reordering.
- **Clinical Quality Monitoring:** using per-test abnormal-result rates to flag tests or conditions that may need closer clinical attention.
- **Business Intelligence:** presenting large, multi-table operational datasets through an interactive dashboard for non-technical decision-makers.

5.3 Social Relevance

Beyond the industry angle, this kind of analysis has a direct social benefit: it helps a hospital use its own data to serve patients better — shorter wait times through better capacity planning, more reliable medicine availability through inventory monitoring, and fairer, more transparent billing practices through financial analysis. Making hospital operations more legible to administrators, without requiring them to read raw spreadsheets or write code, is itself a meaningful accessibility improvement.

5.4 Industry Skills Demonstrated

- Data auditing and quality control
- Data cleaning and validation in Python
- Exploratory Data Analysis
- SQL querying (joins, aggregations, window functions)
- Dashboard development in Power BI
- Cross-tool verification (checking that Python, SQL, and BI results agree)



6. Learning and Reflection

6.1 Technical Learning

- **Python and Pandas:** practical experience loading, cleaning, joining, and analyzing a large multi-table dataset.
- **Data Quality Auditing:** learned that a dataset can look clean at a glance and still hide a structural bug — like a leftover "Total" row — that silently distorts every count downstream unless it's specifically checked for.
- **SQL:** gained practical experience writing joins, aggregations, and window functions, and learned to work around engine-specific gaps (e.g. SQLite has no CORR() or DATEDIFF()).
- **Dashboard design:** learned how to translate a set of numeric findings into a small number of clear visuals that a non-technical stakeholder can read at a glance.
- **Cross-verification:** learned the value of checking the same finding across more than one tool (Python, SQL, Power BI) rather than trusting a single analysis in isolation.

6.2 Project Experience and Reflection

The hardest part of this project was **cleaning the data and connecting the different tables**. There were some missing values, duplicate records, and wrong IDs. It took time to check and fix these issues before doing the analysis.

I was surprised that **length of stay and bill amount had almost no relation**. I thought patients who stayed longer would always have higher bills, but the data did not show this. I also found a difference between the **Excel check for duplicate policy numbers and the SQL check**. This showed me that data should be checked in more than one way.

If I started the project again, I would **clean and check the data first**. I would also plan my dashboards and questions before starting the analysis. This would save time and make the project easier to understand.

6.3 Overall Learning Outcome

This project helped me move from **learning tools separately to using them together**. I used **Python to clean and explore the data, SQL to check and analyze it, and Tableau to create dashboards**. I learned how these tools work together to solve a real problem.

The data was not perfect, so I had to deal with **missing values, duplicate data, and wrong values**. This project taught me that real-world data needs to be checked and cleaned before using it. Overall, I gained confidence in using **Python, SQL, and Tableau together for data analysis**.



7. Conclusion and Future Scope

7.1 Conclusion

The HIMs project set out to turn a raw, 19-table hospital operations workbook into reliable, actionable insight. The project followed a structured process — understanding the data, auditing and cleaning it, analyzing it in both Python and SQL, and presenting it through an interactive Power BI dashboard.

The analysis surfaced a small number of clear, well-supported findings: ICU stays run nearly twice as long as any other department; elective admissions outnumber emergencies roughly 3-to-2; room charges — not procedures or drugs — dominate hospital revenue; about half of all bills remain unpaid; and length of stay has essentially no relationship to how much a patient is billed. Each of these was checked against the real data, not assumed, and several were confirmed independently across Python, SQL, and the Power BI dashboard.

Overall, the project demonstrates a complete, defensible data analytics workflow — from a messy source file to a small set of numbers a hospital administrator could act on.

7.2 Future Scope

- **1. Predictive Analytics:** machine learning could be introduced to predict length of stay from admission details (an initial test showed this is genuinely learnable — department and admission type alone explain roughly 62% of the variance in length of stay) or to flag patients at risk of a long ICU stay before it happens.
- **2. Revenue Collection Analysis:** a deeper look into why ~50% of bills remain pending — by department, payment mode, or patient segment — could turn this from a descriptive number into an actionable collections strategy.
- **3. Real-Time Data Refresh:** connecting the Power BI dashboard to a live data source (rather than a static workbook) so administrators see current, not historical, operations.
- **4. Additional Dashboard Pages:** a dedicated Financial page (payment status/mode, revenue by charge type and department) and a Pharmacy & Staffing page, following the same build guide used for the first three pages.
- **5. Natural-Language Query Interface:** an AI assistant layered on top of the cleaned dataset, letting hospital staff ask questions in plain English instead of writing SQL or navigating dashboard filters.

7.3 Final Statement

This project shows that a raw, imperfect operational dataset can be turned into clear, trustworthy business insight through careful auditing, exploratory analysis, and visualization — and that the same underlying data supports Python, SQL, and BI-tool analysis equally well when it has been cleaned properly first